SUPERSEDING
MIL-N-55491(EL)
16 April 1968

MILITARY SPECIFICATION

NIGHT VISION SIGHT, TRIPOD MOUNTED AN/TVS-4

This specification is approved for use by all Departments and Agencies of the Department of Defense

1. SCOPE

1.1 Scope. This specification covers a night vision sight, for tripod mounting (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

QQ-S-781	Strapping, Steel, Flat and Seals
PPP-B-636	Boxes, Shipping Fiberboard
PPP-F-320	Fiberboard, Corrugated and Solid, Sheet Stock (Container Grade) and Cut Shapes
PPP-P-291	Paperboard, Wrapping and Cushioning
PPP-S-760	Strapping, Nonmetallic, (and Connectors)
PPP-T-76	Tape, Pressure-Sensitive Adhesive Paper, (For Carton Sealing)
PPP-T-97	Tape, Pressure-Sensitive Adhesive, Filament Reinforced

MILITARY

MIL-P-116	Preservation Packaging Methods of
MIL-P-514	Plates, Identification, Instruction and Marking, Blank
MIL-P-11268	Parts, Materials, and Processes Used in Electronic Equipment
MIL-H-15424	Hand Tools, Packaging of
MIL-E-17555	Electronic and Electrical Equipment, Accessories and Repair Parts; Packaging and Packing of

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STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-150	Photographic Lenses
MIL-STD-171	Finishing of Metal and Wood Surfaces
MIL-STD-202	Test Methods for Electronic and Electrical Component Parts
MIL-STD-461	Electromagnetic Interference Characteristics Requirement for Equipment
MIL-STD-705	Generator Sets, Engine Driven Methods of Tests and Instructions
MIL-STD-810	Environmental Test Methods

DRAWINGS

ELECTRONICS COMMAND

SC-DL-627850	Night Vision Sight, Tripod Mounted

(Copies of specifications, standards, and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Description. The night vision sight, tripod mounted (hereinafter referred to as "sight") shall consist of a housing, battery, oscillator (when applicable) yoke, small tripod and carrying case.

3.2 Construction. The sights shall be in accordance with SC-DL-627850 and all provisions of this specification.

3.3 First article. When specified in the contract or purchase order, the contractor shall furnish first article units in accordance with 4.3.

3.4 Parts and materials. Parts and materials shall be as specified herein and as shown on the applicable drawings. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification and shall conform to MIL-P-11268.

3.4.1 Objective lens assembly.

3.4.1.1 Contrast transfer. The contrast transfer for the spectral range of an S-20 or extended S-20 photoemissive surface to a tungsten source

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operated at a color temperature of 2870° K, ± 50° K shall not be less than the following values for the center and edges of the lens assembly for the angular resolutions indicated.

TEST POINT (See Fig 1)	ANGULAR RESOLUTION (LP/Milliradian)	CONTRAST TRANSFER Center Edge (± 3.6°)	
		Tangential	Best Focus
1	0.64	94	92
2	1.88	92	90
3	5.00	74	68

3.4.1.2 Back focal length. The paraxial back focal length (BFL) of the objective lens shall be 6.45 mm, + .20 mm, -1.07 mm (0.262 inch, + .008 inch, -.042 inch).

3.4.1.3 Clear aperture. The clear aperture area of the objective lens shall be 24,750 square mm minimum.

3.4.1.4 Resolution. Resolution at high contrast levels, over the full area of a 20.0 mm radius circle, perpendicular to and coaxial with the optical axis in the focal plane of the objective lens, shall degrade not more than 120 line pairs per mm at the center of 80 line pairs per mm at the edge.

3.4.1.5 Stray light. Stray light luminance contribution over the image of a black disk, subtending a solid angle equal to the field of view of the sights in a white surround, subtending a solid angle of not less than 120° shall not exceed 2.0 percent.

3.4.1.6 Field flatness. The image plane shall be optically flat to a range of 0.1 mm over the full area of a 20 mm radius circle, perpendicular to and coaxial with the optical axis in the image plane when the objective lens is sharply focused for infinity.

3.4.1.7 Color correction. Color correction for the objective lens shall be for the region of spectral response of an extended S-20 photoemissive surface to a tungsten source operated at a color temperature of 2870° K, ± 50° K and shall be subjected to "Color Correction" as specified in MIL-STD-150.

3.4.1.8 Transmission. Transmission shall be not less than 70.0 percent over the full objective lens aperture, for the region of spectral response of an S-20 or extended S-20 photoemissive surface to a tungsten source operated at a color temperature of 2870° K, ± 50° K.

3.4.2 Eyepiece lens assembly.

3.4.2.1 Eye excursion, field coverage and eye relief distance. The eyepiece shall provide full field coverage equal to the total solid angle subtended by a 19 mm radius circle perpendicular to and coaxial with the optical axis in the focal plane. The vignetting at the edge of the field

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shall be controlled so that none occurs for eye excursion over the full area of a 7.0 mm diameter aperture which is perpendicular to and coaxial with the optical axis at an eye relief distance of 25.4 mm when the eyepiece is sharply focused and once focused not further changed.

3.4.2.2 Contrast transfer. The contrast transfer for the region of spectral response to a P-20 phosphor of an eye corrected detector shall be not less than the following values for the center of the field of view for angular resolutions indicated when tested through circular apertures located at the eye relief distance.

TEST POINT (See Fig 1)	ANGULAR RESOLUTION (LP/Millian)	7 mm APERTURE Contrast Transfer Edge (±24°) Tangential Best Focus	15 mm APERTURE Contrast Transfer Center
1	0.09	0.90	0.95
2	0.28	0.85	0.90
3	0.74	0.55	0.80
4	1.12	0.50	0.65

3.4.2.3 Resolution. The resolution through a 15 mm aperture for the region of spectral response of a P-20 phosphor shall be not less than 80 line pairs per mm in the center of the field, and shall degrade to not less than 35 line pairs per mm throughout a 19 mm radius circular field of view for radial and tangential lines.

3.4.2.4 Field flatness. The image plane as observed through a 15 mm aperture shall be optically flat within a range of 0.5 diopter over the field specified in 3.4.2.1.

3.4.2.5 Color correction Color correction for the eyepiece shall be for the region of spectral response for an eye corrected P-20 phosphor, and shall be subject to the provisions of "Color Correction" as specified in MIL-STD-150.

3.4.2.6 Transmission. Transmission shall be not less than 80 percent for the region of spectral response of an eye corrected P-20 phosphor.

3.4.2.7 Equivalent focal length (EFL). The EFL of the eyepiece as tested through a 15 mm circular aperture located at the eye relief distance shall be 37.4 mm, plus or minus 0.8 mm.

3.5 Operational and environmental.

3.5.1 Relative humidity. Humidity of 90 to 98 percent at an ambient temperature of from 29°C to 49°C, plus or minus 3°C.

3.5.2 Thermal shock, operation and high and low temperature storage.

a. Temperature shocks from plus 46°C to plus 21°C, plus or minus 3°C, and from minus 54°C to plus 21°C, plus or minus 3°C.

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b. Storage at an ambient temperature of plus 68°C to plus 21°C, plus or minus 3°C and from minus 54°C for two periods of 12 hours per day.

c. Operation at plus 46°C and minus 54°C.

3.5.3 Barometric pressure (reduced). The sight shall operate as specified herein and shall not be damaged at an latitude of 40,000 feet (5.5 inches of mercury pressure).

3.5.4 Salt fog. The sights shall operate as specified herein, and shall nt be damaged after having been tested in accordance with Method 509.1 of MIL-STD-810.

3.5.5 Immersion. The sight shall operate as specified herein, and shall not be damaged after immersion to a depth of 3 feet for 30 minutes in fresh water.

3.5.6 Electromagnetic interference. The electromagnetic interference of the sight shall be in accordance with MIL-STD-461, Table A-1, MIL-E-55301, Equipment Class-Tactical-Non-CE.

3.5.7 Fungus. The equipment (including accessories) shall show no evidence of viable fungus or corrosion when subjected to the test specified in 4.6.7. Corrosion is any visible degradation that can be attributed to flaky, pitted, blistered, or otherwise loosened finish or metal surface.

3.5.8 Ruggedness. The sight shall not be damaged (see 6.3.1) after having been tested for 30 minutes in accordance with MIL-STD-705, Method 740.1a, except that test method 614.1 shall not apply.

3.5.9 Vibration. The sight shall not be damaged and shall operate as specified herein after being subjected to vibration of 10 to 55 hertz (Hz) in three mutually perpendicular planes, one plane being parallel to the optical axis. The sights shall have no resonant frequency below 20 Hz.

3.5.10 Resolution. The sight shall resolve 3.8 line pairs per milliradian on-axis and 1.9 line pairs per milliradian at 3.6° off-axis without refocusing of the objective lens for the spectral range of a tungsten source operated at a color temperature of 2870° K, ± 50° K when tested through a 7 millimeter (mm) circular aperture located at the eye relief distance of 25.4 mm.

3.5.11 Magnification. The magnification at 3.6° from the center of the field of view shall be not less than 7.0 power (X) nor greater than 7.7 X.

3.5.12 Field of view. The field of view shall be not less than 8.0°.

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3.5.13 Focus adjustment. The field of view shall be not less than the objective and eyepiece lenses shall require:

a. Objective focus from 50 meters to infinity by applying not more than 15 inch-pounds of running torque.

b. Eyepiece focus of plus 4 to minus 3.25 diopters, plus or minus 0.25 diopter by applying not more than 30 inch-pounds of running torque with backlash not exceeding 5° rotation in either direction with eyepiece properly installed in a complete sight (including an image intensifier assembly).

3.6 Government-loaned property. The following property in the quantities indicated, will be furnished by the Government when specified in the contract (see 6.2).

ITEM	DESCRIPTION	IDENTIFICATION	QUANTITY
1	Image Intensifier Assembly (40MM)	NSN 5855-00-908-9314 or NSN 5855-00-147-2508	6
2	Dummy Image Intensifier (40MM)		6

3.7 Identification marking. Identification markings shall be in accordance with the applicable drawings and MIL-STD-130. Plates shall conform to MIL-P-514, Type III, Composition.

3.8 Treatment and painting. The portions of the sight and its components and parts normally painted shall be cleaned, treated and painted in accordance with the applicable part drawing and in accordance with MIL-STD-171.

3.9 Workmanship. All parts, components and assemblies of the sight including castings, forgings, molded parts, stampings, bearings, sand, fins, pits, spruces, scale, flux and other harmful extraneous materials. Joints and seams shall be tight fit, all edges shall be smoothly finished. The interior of the AN/TVS-4 shall be free of all foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Parts and materials. The supplier is responsible for insuring that parts and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards as specified herein.

4.2 Classification of inspection. Inspection shall be classified as follows:

a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

c. Inspection of preparation for delivery (see 4.7).

4.3 First article inspection. Unless otherwise specified in the contract or purchase order, the first article inspection shall be performed by the contractor.

4.3.1 Inspection. Each sight shall be examined in accordance with Table I. One or more defects shall be cause for rejection of that sight.

TABLE I Inspections

Defects	Requirement paragraph	Inprocess	Assembled
MAJOR			
101. Parts and materials missing or not as specified	3.4	X	
102. Dimension not as specified	3.2	X	
103. Identification or special marking missing or illegible	3.2	X	
104. Workmanship not as specified	3.9	X	
105. Controls move freely without binding	3.5.13		X
106. Diopter focus gives sharp image of infinity target	3.5.13		X

4.3.2 Tests. The first article sight shall be subjected to the tests in Table II. Tests shall be conducted in the order listed. Failure of any test shall be cause for rejection.

TABLE II First article inspection

Inspection	Requirement paragraph	Test paragraph
Objective assembly:		
Contrast transfer	3.4.1.1	4.5.1.1
Back focal length	3.4.1.2	4.5.1.2
Clear aperture	3.4.1.3	4.5.1.3
Resolution	3.4.1.4	4.5.1.4
Stray light	3.4.1.5	4.5.1.5

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TABLE II First article inspection (con't)

Inspection	Requirement paragraph	Test paragraph
Field flatness	3.4.1.6	4.5.1.6
Color correction	3.4.1.7	4.5.1.7
Transmission	3.4.1.8	4.5.1.8
Eyepiece lens assembly:		
Eye excursion, field coverage eye relief distance	3.4.2.1	4.5.2.1
Contrast transfer	3.4.2.2	4.5.2.2
Resolution	3.4.2.3	4.5.2.3
Field flatness	3.4.2.4	4.5.2.4
Color correction	3.4.2.5	4.5.2.5
Transmission	3.4.2.6	4.5.2.6
Equivalent focal length	3.4.2.7	4.5.2.7
Sight assembly:		
Magnification	3.5.11	4.6.11
Resolution	3.5.10	4.6.10
Field of view	3.5.12	4.6.12
Focus adjustment	3.5.13	4.6.13
Ruggedness	3.5.8	4.6.8
Vibration	3.5.9	4.6.9
Resolution	3.5.10	4.6.10
Focus adjustment	3.5.13	4.6.13
Relative humidity	3.5.1	4.6.1
Thermal shock, operation and high and low temperature storage	3.5.2	4.6.2
Resolution	3.5.10	4.6.10
Focus adjustment	3.5.13	4.6.13
Barometric pressure (reduced)	3.5.3	4.6.3
Salt fog	3.5.4	4.6.4
Fungus	3.5.7	4.6.7
Immersion	3.5.5	4.6.5
Resolution	3.5.10	4.6.10
Focus adjustment	3.5.13	4.6.13
Electromagnetic interference	3.5.6	4.6.6

4.4 Quality conformance inspection.

4.4.1 Inspection. Each sight shall be inspected as specified in 4.3.1. Presence of one or more defects shall be cause for rejection of that unit. Examination will be performed upon the lot after successful completion of Group A and B inspection.

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4.4.2 Group A Inspection. Each sight shall be inspected in accordance with Table III. Failure of any test shall be cause for rejection.

TABLE III Group A Inspection

Inspection	Requirement paragraph	Test paragraph
Objective lens assembly:		
Contrast transfer, on-axis only	3.4.1.1	4.5.1.1
Back focal length	3.4.1.2	4.5.1.2
Eyepiece lens assembly:		
Contrast transfer on-axis only	3.4.2.2	4.5.2.2

4.4.3 Group B Inspection. Group B inspection shall be conducted on units which have passed tests specified in 4.4.1. Sample size shall be in accordance with MIL-STD-105, Inspection Level S-3, AQL 1.5 percent defective. Tests shall be conducted in accordance with Table IV, and may be performed in any sequence except the vibration test shall be first followed by the immersion test. Failure of any test shall be cause for rejection of the lot.

TABLE IV Group B Inspection

Inspection	Requirement paragraph	Test paragraph
Sight assembly:		
Vibration	3.5.9	4.6.9
Immersion	3.5.5	4.6.5
Focus adjustment	3.5.13	4.6.13
Objective lens assembly:		
Contrast transfer, off-axis	3.4.1.1	4.5.1.1
Transmission	3.4.1.8	4.5.1.8
Eyepiece lens assembly;		
Contrast transfer, off-axis	3.4.2.2	4.5.2.2

4.4.4 Group C Inspection. This inspection shall consist of the tests specified in Table V and shall be performed on units that have been subjected to and which have met Group A inspection. Units shall be selected in accordance with 4.4.4.1. The tests may be performed in any order except that the two contrast transfer tests shall be last.

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4.4.4.1 Sampling for inspection. Two units shall be selected in accordance with 4.4.4.2 by the Government for all inspections in Table V whenever the monthly quantity specified in the contract starts quality conformance testing or every 30 days whichever occurs first. The units are not associated with inspection lots as prescribed by MIL-STD-105, paragraph 5.1. The only selection prerequisite is that the units conform to normal production processing. The first Group C units shall begin test at the time the first quality conformance lot (see 4.4.3) begins Group B inspection.

4.4.4.2 Method of sampling. The number of units to receive Group C inspection per month is as follows:

Number of units w/o intensifier to receive Group C inspection	Quantity of units w/o intensifier being delivered within a month
2	0 to 100
3	101 to 150
4	151 to 200
6	201 to 300
8	301 to 400

TABLE V Group C Inspection

Inspection	Requirement paragraph	Test paragraph
Ruggedness	3.5.8	4.6.8
Relative humidity	3.5.1	4.6.1
Thermal shock, operation and high and low temperature storage	3.5.2	4.6.2
Resolution	3.5.10	4.6.10
Field of view	3.5.12	4.6.12
Focus adjustment	3.5.13	4.6.13
Objective contrast transfer, off-axis, at 5 LP/MR only	3.4.1.1	4.5.1.1
Field flatness	3.4.1.6	4.5.1.6
Eyepiece contrast transfer, off-axis, at 1.12 LP/MR only	3.4.2.2	4.5.2.2
Field flatness	3.4.2.4	4.5.2.4

4.4.4.3 Group C failures. Actions required relative to Group C failures shall be as specified in the contract or purchase order (see 6.2).

4.5 Test methods. Unless otherwise specified, all tests shall be performed at plus 21°C plus or minus 3°C. The ruggedness test in Table II and Table V shall be conducted on the sight described in 3.1. Unless otherwise specified herein all other tests shall be conducted on the sight less the yoke, tripod and carrying case.

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4.5.1 Objective lens assembly.

4.5.1.1 Contrast transfer. Place the objective lens assembly between the collimator and S-20 photomultiplier tube or between a coupled extended S-20 image intensifier assembly and a S-11 photomultiplier tube as shown in Figure 5. Illuminate the frequency pattern with light from a 2870° K, ± 50° K tungsten lamp and focus the image of the frequency pattern at the slit with a microscope objective. Adjust the X-Y recorder for zero level by blocking the entrance of all light to the photo-detector. Advance the frequency pattern (see Figure 1A or 1B), through any three of the four test point frequencies which subtend the required LP/milliradian and obtain recordings for the center and edge of the objective lens (see Figure 6). Calculate the contrast transfer from the measured data using the formula: $\frac{W-B}{W+B}$.

NOTE: Failure to comply with the values specified in 3.4.1.1 shall constitute failure of this test. Special care must be taken before test data is recorded to avoid the possible introduction of contrast transfer losses resulting from the test instrument.

4.5.1.2 Back focal length. The BFL of the objective lens assembly shall be measured in accordance with MIL-STD-150. Failure to comply with the BFL requirement of 3.4.1.2 shall constitute failure of the test.

4.5.1.3 Clear aperture. The clear aperture of the objective lens assembly shall be measured in accordance with effective aperture methods of MIL-STD-150. Failure to produce the clear aperture specified in 3.4.1.3 shall constitute failure of this test.

4.5.1.4 Resolution. The resolution of the objective lens shall be determined in accordance with the method stated under "resolving power" of MIL-STD-150. Failure to comply with the resolution values given in 3.4.1.4 shall constitute failure of this test.

4.5.1.5 Stray light. The test apparatus shown in Figure 7 shall be used to measure the stray light luminance. The illuminated field shall subtend a solid angle of not less than 120° centered in respect to the optical axis and shall provide best possible uniform brightness. The objective shade shall be placed on the objective and extended sufficiently to prevent any external light from reaching the focal surface by going directly past the secondary. The objective is centrally located at the stray light fixture aperture. A diffusion plate is placed over the image plane of the objective and the stray light-fixture turned on. The photometer is focused centrally on the diffusion plate and the output reading adjusted to 100 percent. A circular obscuration disc of sufficient diameter as to subtend an angle of 10°, plus or minus 30 minutes plus the clear aperture of the objective is placed centrally on the axis of the objective and on the illuminated field of the stray light fixture. A second reading is taken with the photometer focused on the center of the dark area of the diffusion plate in the objective image plane. The obscured reading shall be no more than 1.3 percent of the unobscured field reading. A reading in excess of 1.3 percent shall constitute failure of this test.

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4.5.1.6 Field flatness. The field flatness of the objective lens shall be measured in accordance with the astigmatism and curvature of field test methods of MIL-STD-150. Failure to comply with the values specified in 3.4.1.6 shall constitute failure of this test.

4.5.1.7 Color correction. Color correction shall be measured as specified in MIL-STD-150. Failure to comply with the requirements of 3.4.1.7 shall constitute failure of this test.

4.5.1.8 Transmission. The transmission of the objective lens shall be measured in accordance with "T-number and the transmittance" of MIL-STD-150. When a suitable sensor with extended S-20 response is not available, a sensor with visual response may be employed. Failure to comply with the values specified in 3.6.8 shall constitute failure of this test.

4.5.2 Eyepiece lens assembly.

4.5.2.1 Eye excursion, field coverage and eye relief distance. Observe the 40 m m circle through the 15 mm aperture as shown in Figure 8. Move the eye from one edge of the aperture to the other. Loss of field or vignetting over the 40 mm circle shall constitute failure of this test.

4.5.2.2 Contrast transfer. Place the eyepiece lens assembly between the collimator and S-20 photomultiplier tube or a coupled extended S-20 image intensifier tube and S-11 photomultiplier tube as shown in Figure 5. Illuminate the frequency pattern with light from a 2870° K, ± 50° K tunsten lamp and focus the image of the frequency pattern at the slit with a microscope objective. Adjust the X-Y recorder for zero level by blocking the entrance of all light to the photo-detector. Advance the frequency pattern (see Figure 1A or 1B), through the four test point frequencies and obtain recordings for the center and edge of the eyepiece lens (see Figure 6). Calculate the contrast transfer from the measured data using the formula: $\frac{W-B}{W+B}$.
Failure to meet requirements of 3.4.2.2 shall constitute failure of this test.

4.5.2.3 Resolution. With the aid of the test apparatus shown in Figure 8, the on-axis resolution of the eyepiece shall be determined in accordance with the "resolving power" of MIL-STD-150. The off-axis radial and tangential resolution shall be determined at 4 lines respectively. Failure to comply with the resolutions values specified in 3.4.2.3 shall constitute failure of this test.

4.5.2.4 Field flatness. The field flatness of the eyepiece as determined through the 15 mm aperture as shown in Figure 8 shall be measured in accordance with the "astigmatism and curvature of field" test methods of MIL-STD-150. Failure to comply with the values specified in 3.4.2.4 shall constitute failure of this test.

4.5.2.5 Color correction. Color correction shall be measured as specified in MIL-STD-150. Failure to comply with the requirements of 3.4.2.5 shall constitute failure of this test.

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4.5.2.6 Transmission. The transmission of the eyepiece as determined through a 15 mm aperture located at the eye relief distance shall be measured in accordance with "T-number and transmittance" of MIL-STD-150. Failure to comply with the values specified in 3.4.2.6 shall constitute failure of this test.

4.5.2.7 Equivalent focal length. The equivalent focal length of the eyepiece shall be measured in accordance with MIL-STD-150. Failure to comply with the EFL specified in 3.4.2.7 shall constitute failure of this test.

4.6 Operation and environmental.

4.6.1 Relative humidity. Place the sight in a chamber with an internal temperature of 29°C and a relative humidity of 90 to 98 percent and soak for 4 hours. Raise the temperature of the chamber to 49°C, plus or minus 3°C, in not more than 2 hours, maintain the relative humidity of 90 to 98 percent and soak for 4 hours. Lower the temperature of the chamber to 29°C in not more than 2 hours. This constitutes one complete cycle as shown in Figure 3. Perform the above cycle six times. Turn the sight ON and observe for image tube operation for the last 30 minutes of the 29°C portion of the fourth and sixth cycles. Failure to operate or evidence of damage during, or immediately following the above six cycles shall constitute failure of this test.

4.6.2 Thermal shock, operation, high and low temperature storage. Place the sight in a chamber which has an internal temperature of 85°C and soak for 4 hours. Lower the temperature in the chamber to 46°C in not more than 2 hours and soak for 4 hours. Turn the sight ON and observe for image intensifier operation during the first 20 minutes of this 4-hour period. Remove the sight from the chamber and soak for 2 hours at 21°C Place the sight in a chamber which has an internal temperature of minus 54°C. Soak the sight at this temperature for 12 hours. Turn the sight ON and observe for image intensifier operation for the last 30 minutes of the twelfth hour. It is permissible to replace the battery in the unit at this time, with a battery which has been at 21°C in order to operate this unit. Remove the viewer from the chamber and soak for 4 hours at 21°C. Turn the viewer ON and observe for image intensifier operation for the last 10 minutes of this period. This constitutes one complete 28 hour cycle as shown in Figure 4. Repeat the above cycle. Failure to operate or evidence of damage during or immediately following the above two cycles shall constitute failure of this test.

4.6.3 Barometric pressure (reduced). Test the sight in accordance with Method 105-C of MIL-STD-202 at an altitude of 40,000 feet (5.5 inches of mercury pressure) for 12 hours mounted horizontally. At the end of this period increase the barometric pressure to the ambient, remove the sight from the chamber and operate the sight. Failure to operate as specified in 3.5.3 or evidence of any damage shall constitute failure of this test.

4.6.4 Salt fog. Test the sight in accordance with Method 509-1 of MIL-STD-810, except that a 20 percent salt solution shall be used.

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4.6.5 Immersion. Immerse the sights to a depth of 3 feet in fresh water having a temperature of plus 29°C to plus 35°C for a period of 30 minutes. Upon removal, dry all external surfaces and inspect for entrance of water or moisture. Upon completion of this examination, turn the sight ON and observe for image intensifier assembly operation for 20 minutes. Failure to operate or evidence of leakage or other damage shall constitute failure of this test.

4.6.6 Electromagnetic interference. The first article unit equipped for the reduction of radio interference as specified in 3.5.6 shall be tested as specified herein. The supplier shall furnish the contracting officer the reports required by MIL-STD-461, except that the time interval for submission of reports shall be as specified (see 6.5). Upon approval of the reports by the contracting officer and provided all other requirements of the specification are met, the first article unit shall be used as a model for all production sights.

4.6.7 Fungus. The sight (including accessories) shall be tested in accordance with Method 508, Procedure 1, of MIL-STD-810. Operation is not required.

4.6.8 Ruggedness. Test the sight in accordance with MIL-STD-705, Method 740.1 with a vibration time of 30 minutes, except that steps a and e of the procedure shall be omitted. The sight shall be packed in its case and the case positioned on its bottom. Evidence of damage (see 6.3.1) shall constitute failure of this test.

4.6.9 Vibration. The vibration test shall be conducted using a dummy image intensifier assembly, 40 MM. During the operational inspection, following the test, an operational image intensifier assembly, 40 MM shall be used. Mount the sight less the objective shade rigidly to a vibration table in a horizontal plane and vibrate the sight at an amplitude of 0.05 inch (total excursion 0.10 inch) at nonresonant frequencies between 10 and 55 Hz for a period of 5 minutes in each of the following directions:

a. Perpendicular to the optical axis of the objective lens.

b. Perpendicular to the optical axis of the objective lens, but displaced 90° from a above.

c. Parallel to the optical axis of the objective lens.

Inspect the sight for operational or physical damage resulting from the vibration. Evidence of damage shall constitute failure of this test.

4.6.10 Resolution. The apparatus shown in Figure 2 shall be used to measure the resolution on-axis and 3.6° off-axis. The light shall be a 2870°K, ± 50°K tungsten lamp. Adjust the objective lens of the sight at infinity focus and do not change it again until both on-axis and off-axis resolutions have been measured. Measure on-axis resolution. Rotate the rotary table 3.6° degrees off-axis and measure off-axis resolution. Failure to comply with the resolution specified in 3.5.10 shall constitute failure of this test.

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4.6.11 Magnification. The apparatus shown in Figure 2 shall be used to measure the magnification. With the sight in place on-axis as shown in Figure 2, view the center of the target pattern through the sight with a telemicroscope. Rotate the sight through an angle θ equal to 3.6°. Measure the angle which the exit beam makes with the optical axis. The magnification shall be calculated as tan a/tan. Failure to comply with the magnification specified in 3.5.11 shall constitute failure of this test.

4.6.12 Field of view. The test apparatus shown in Figure 2 shall be used to project a test pattern onto the objective lens of the sight. The sight shall then be pivoted either to the left or right until the center point of the test pattern seen through the sight is at the edge of the field of view. The sight shall then be pivoted in the opposite direction until the center point of the test pattern is at the opposite edge of the field of view. The total angle through which the sight was pivoted shall be measured. A field of view less than 8.0° shall constitute failure of this test.

4.6.13 Focus adjustment.

4.6.13.1 Objective lens. The sight objective lens shall be focused on an object by viewing a real object or a collimated image which is at infinity by focusing the image intensifier assembly. Then place a second object at an optical distance not greater than 50 meters, using a defocused collimated image if desired, and view through the sight by adjusting the objective lens to a position of sharp focus. Failure to focus objects at infinity or 50 meters shall constitute failure of this test.

4.6.13.2 Eyepiece lens. The sight eyepiece lens shall be focused on any object by viewing a real or collimated image. The required stop shall be placed at the eye relief distance, and the eyepiece adjusted to an extreme focusing position. A diopter tester shall be placed as close to the eyepiece lens as possible, and adjusted until the object viewed through the sight and diopter tester combination is in clear focus. The above procedure shall be repeated after adjusting the eyepiece for the other extreme position. Failure of the eyepiece to be adjustable as specified in 3.5.13 shall constitute failure of this test.

4.7 Inspection of preparation for delivery. Preparation for delivery shall be inspected in accordance with MIL-P-116 to determine compliance to requirements of Section 5.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C, as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Cleaning. Cleaning shall be accomplished in accordance with process C-1 of MIL-P-116.

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5.1.1.2 Drying. Drying shall be accomplished in accordance with the applicable procedure of MIL-P-116.

5.1.1.3 Unit packaging. Unit packaging shall be in accordance with methods prescribed in MIL-P-116 and as specified herein.

5.1.1.3.1 Night vision sight, tripod mounted. Night vision sight, tripod mounted shall be preserved and packaged in accordance with Method III of MIL-P-116 as follows:

5.1.1.3.2 Hand tools. Hand tools, as required, shall be packaged in accordance with the applicable requirements of MIL-H-15424.

5.1.1.3.3 Components. Components, as required, shall be packaged in accordance with the applicable requirements of MIL-E-17555.

5.1.1.3.4 Hardware. Hardware, as required, shall be packaged in accordance with the applicable requirements of MIL-P-116. Like stock numbered hardware items shall be packaged and grouped together.

5.1.1.3.5 Consolidation. Consolidate the components packaged as specified above in accordance with 5.1.1.3.5.1 or 5.1.1.3.5.2, as applicable.

5.1.1.3.5.1 Stowage within case. Stow the packaged components within the case to the extent practicable. Fill all voids with paperboard conforming to PPP-P-291, Type III, to prevent movement. The balance of the packaged components, if any, shall be consolidated within a close-fitting fiberboard box conforming to PPP-B-636, type CF, class weather-resistant. Fill all voids with paperboard conforming to PPP-P-291, type III to prevent movement. Closure shall be in accordance with the appendix of the box specification. Cushion the case on all surfaces with cells or pads or both, fabricated of fiberboard conforming to PPP-F-320, type CF, class weather-resistant. Place the cushioned case and consolidating box, if any, within a close-fitting fiberboard box conforming to PPP-B-636, type CF, class weather-resistant. Closure shall be in accordance with the appendix of the box specification. In addition, all seams and joints shall be sealed with tape, not less than 2 inches wide, conforming to PPP-T-76.

5.1.1.3.5.2 Consolidating package. Consolidate the packaged components within a close-fitting fiberboard box conforming to PPP-B-636, type CF, class weather-resistant. Fill all voids with paperboard conforming to PPP-P-291, type III, to prevent movement. Closure shall be in accordance with the appendix of the box specification. In addition, all seams and joints shall be sealed with tape, not less than 2 inches wide, conforming to PPP-T-76.

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5.1.2 Level C. Each night vision sight, tripod mounted shall be preserved and packaged in a manner that will afford adequate protection against corrosion, deterioration and damage during shipment from the supply source to the first receiving activity.

5.2 Packing. Packing shall be level A, B or C as specified.

5.2.1 Level A. Night vision sight, tripod mounted packaged as specified in 5.1, shall be packed in containers conforming to PPP-B-636, type CF, class weather-resistant.

5.2.1.1 Container closure. Closure shall be in accordance with the applicable container specification or appendix thereto.

5.2.1.2 Strapping and reinforcing. Metal strapping conforming to QQ-S-781, type 1, class B shall be applied to wood and wood-cleated shipping containers in accordance with the requirements of the applicable box specification or appendix thereto. Fiberboard shipping containers shall be reinforced by pressure sensitive filament tape banding or non-metallic strapping conforming to PPP-T-97, type IV, and PPP-S-760, type II respectively; selection of the material and application shall be in accordance with the appendix of the applicable box specification.

5.2.2 Level C. Night vision sight, tripod mounted shall be packed in shipping containers in a manner that will afford adequate protection against damage to the package and its contents during shipment from the supply source to the first receiving activity. Shipping containers shall comply with the rules and regulations of the common carrier as applicable to the mode of transportation.

5.3 Marking. In addition to any special marking required by the contract or order, interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The TVS-4 is intended to provide visibility in the dark using only ambient light.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.

b. Government loaned property (see 3.6).

c. Delivery schedule of first article (see 3.3).

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d. Necessary actions by the contractor in the event of a Group C failure (see 4.4.4.3).

e. Level of packaging, packing required (see 5.).

f. State time interval for submitting reports on Electromagnetic Interference (see 4.6.6).

6.3 Definitions.

6.3.1 Damage. Any breakage, loosening, failure, deterioration or corrosion of any finish, hardware, optical coating, connection or component. Leakage or condensation of moisture into or within the sight.

6.3.2 Room temperature. Room temeprature is defined as plus 23°C, plus or minus 3°C.

6.4 Government loaned property. The contracting officer shall arrange to furnish the property specified in 3.6).

6.5 Electromagnetic interference. The contracting officer shall forward a copy of the test report (see 4.6.6) to the Commander, US Army Electronics Command, ATTN: AMSEL-RD-EB, Fort Monmouth, New Jersey 07703.

CUSTODIAN:

Army-EL
Navy-EC
Air Force-84

USER ACTIVITY:
Army-MU,ME
Navy-OS

PREPARING ACTIVITY:

Army-EL

Project No. 5855-0018

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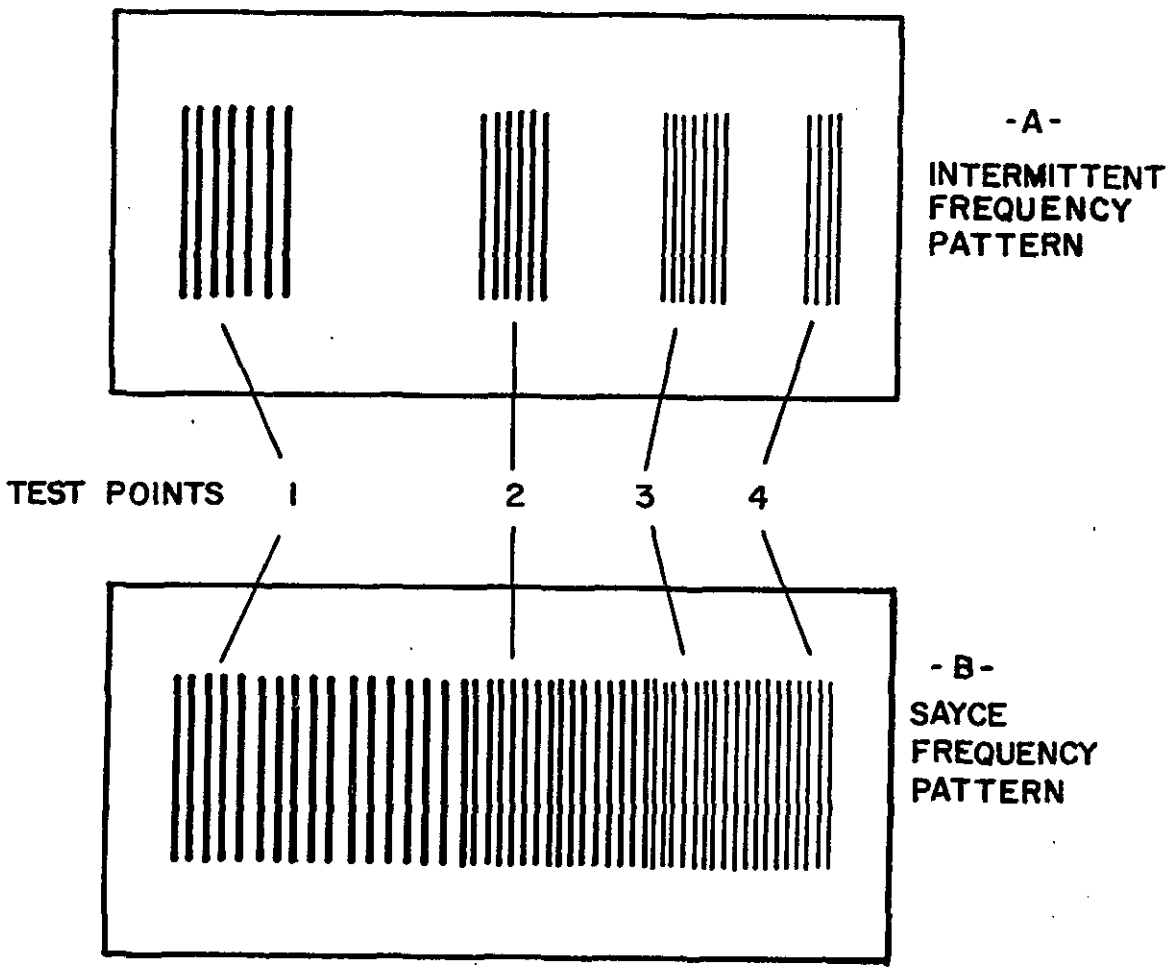


FIGURE 1. FREQUENCY PATTERNS FOR CONTRAST TRANSFER TEST

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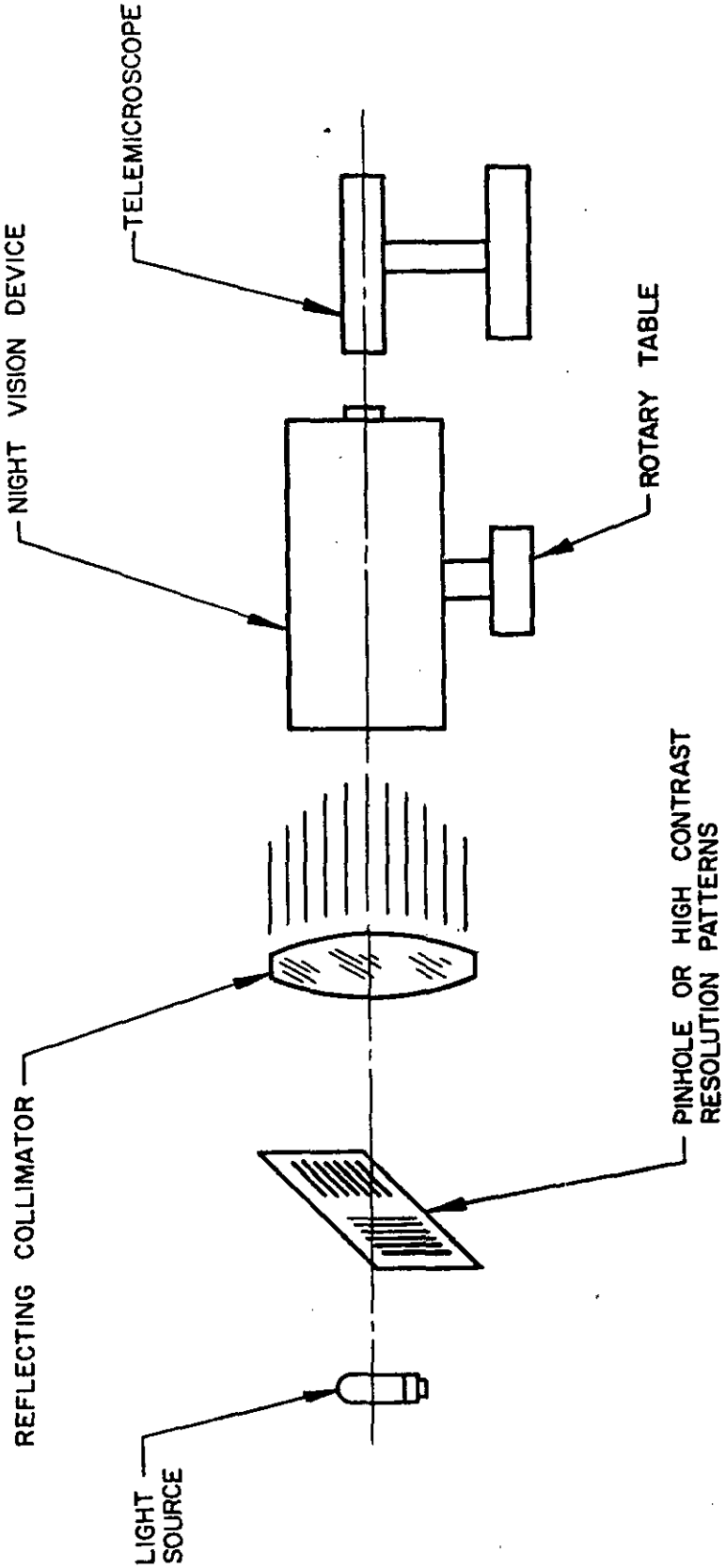


FIGURE 2

PERIPHERAL MAGNIFICATION, RESOLUTION AND FIELD OF VIEW TEST SCHEMATIC

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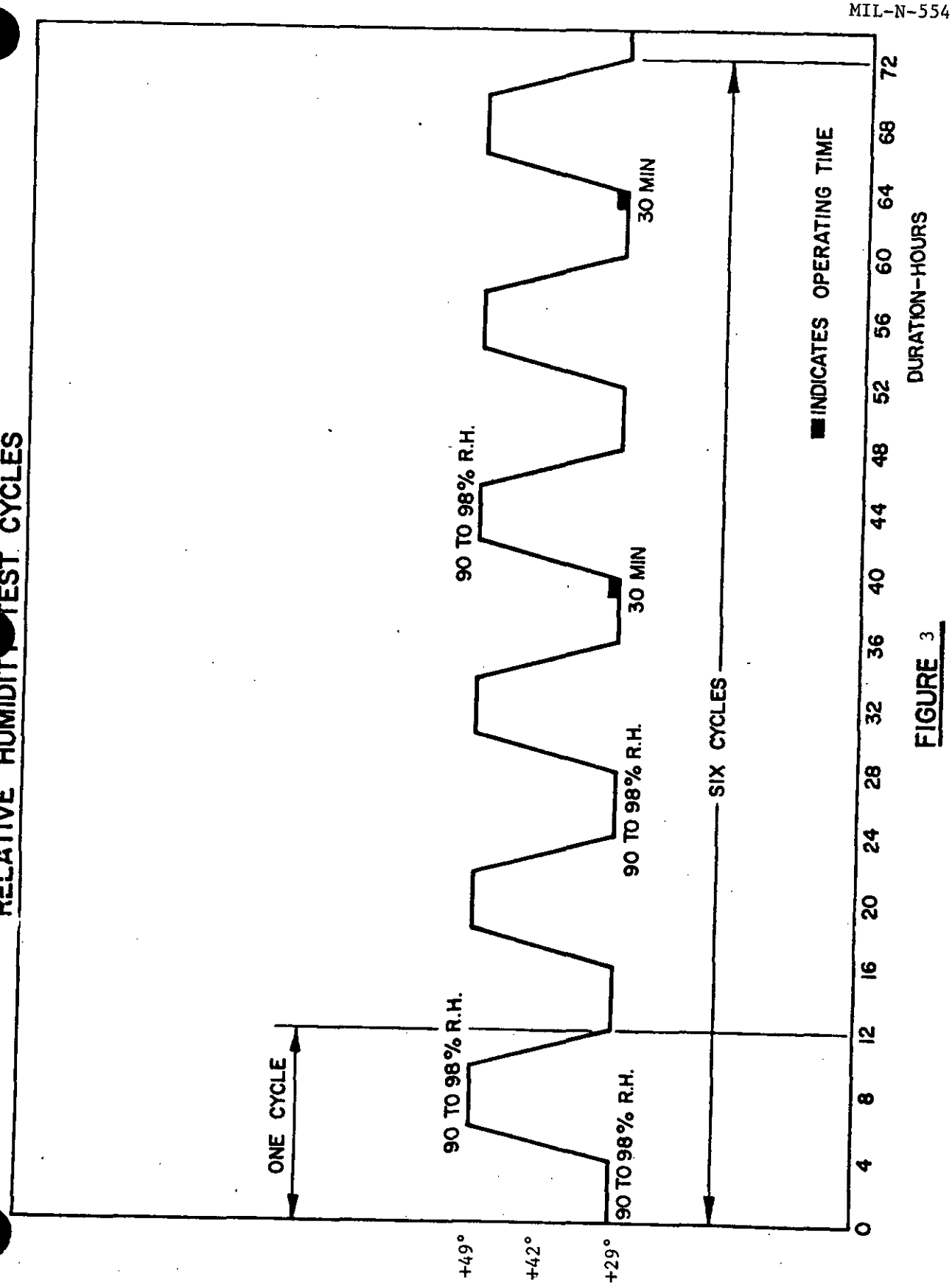


FIGURE 3

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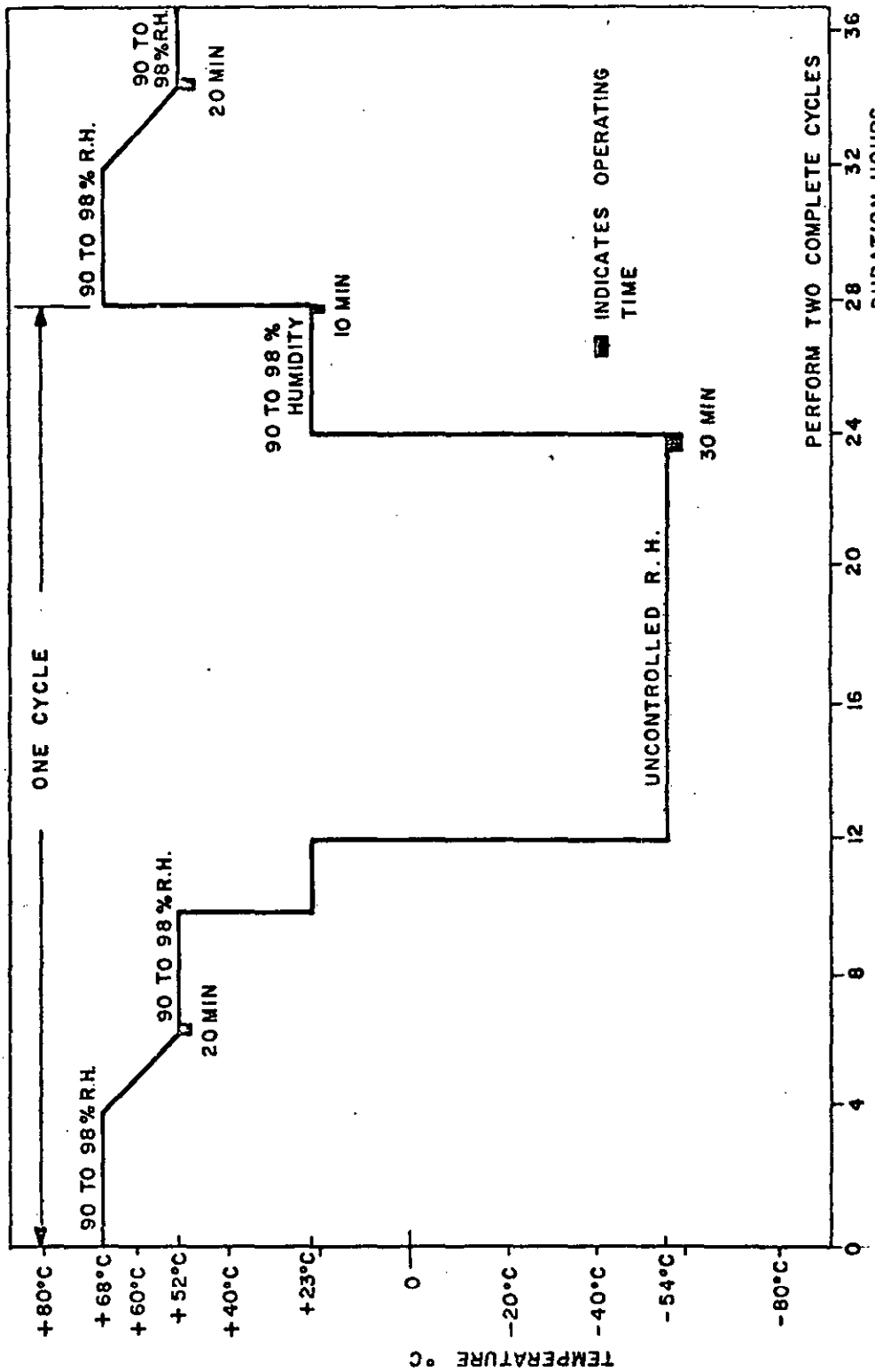


FIGURE 4. STORAGE, OPERATION AND THERMAL SHOCK

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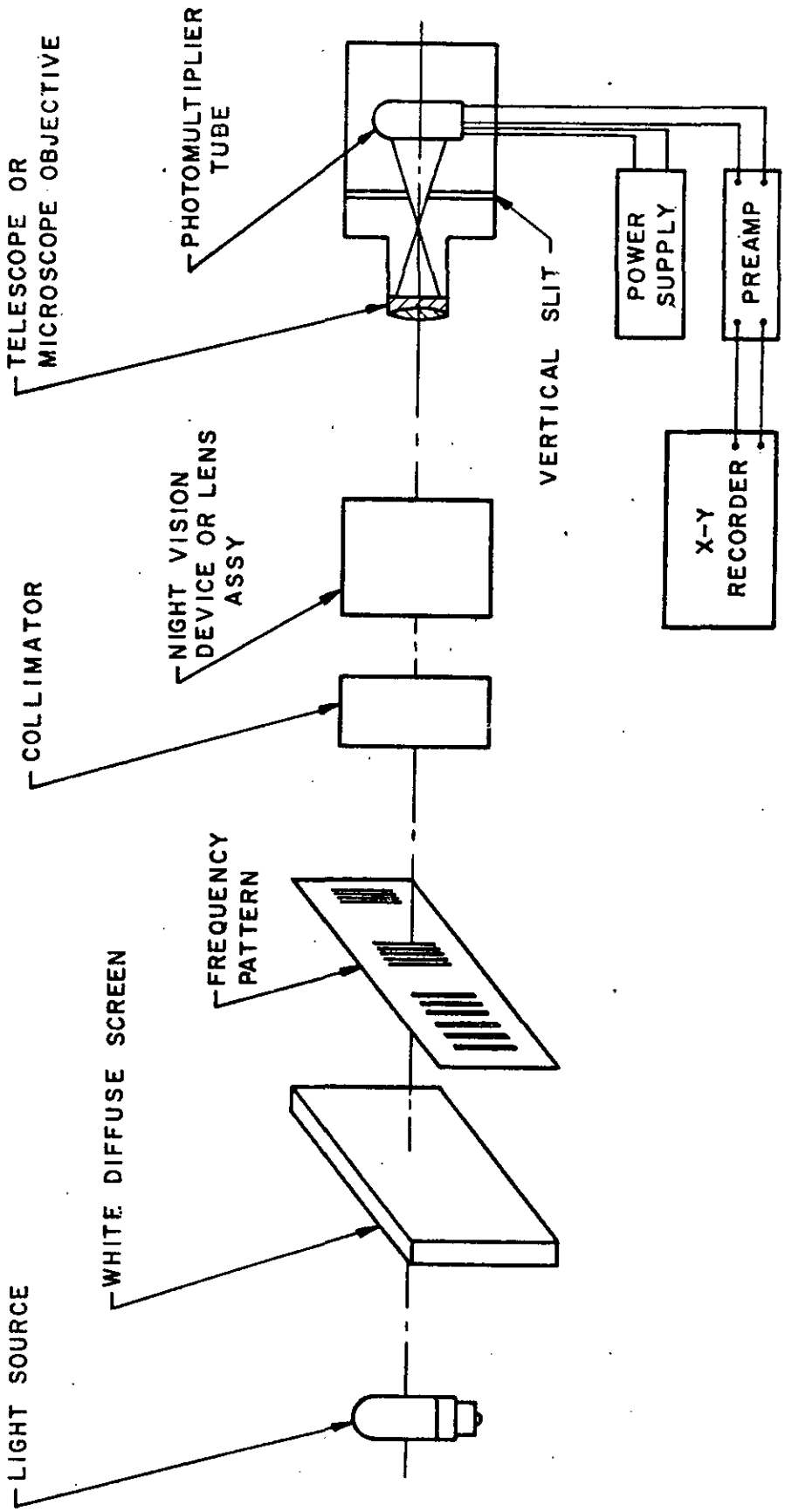
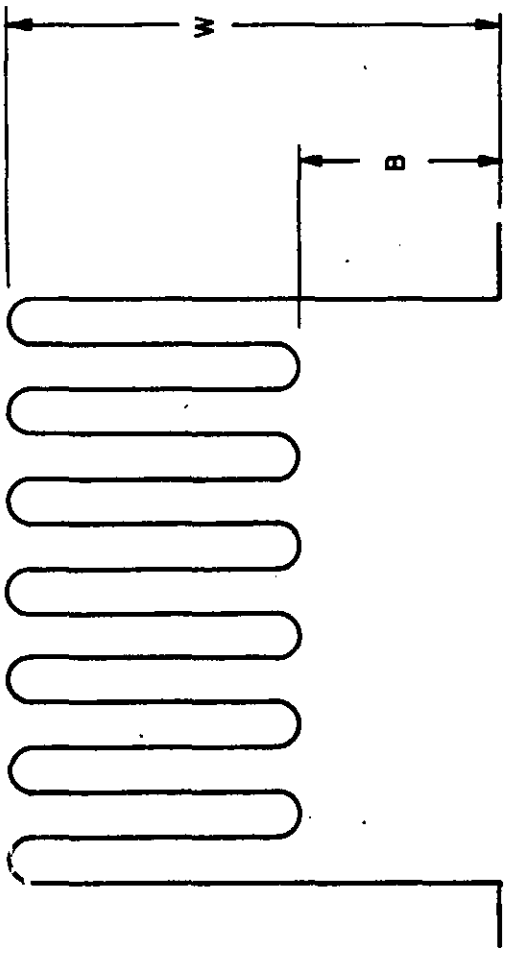


FIGURE 5.

CONTRAST TRANSFER TEST APPARATUS

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NOTE: CONTRAST TRANSFER IS DEFINED AS $\frac{W-B}{W+B}$

FIGURE 6

CONTRAST TRANSFER WAVE FORM

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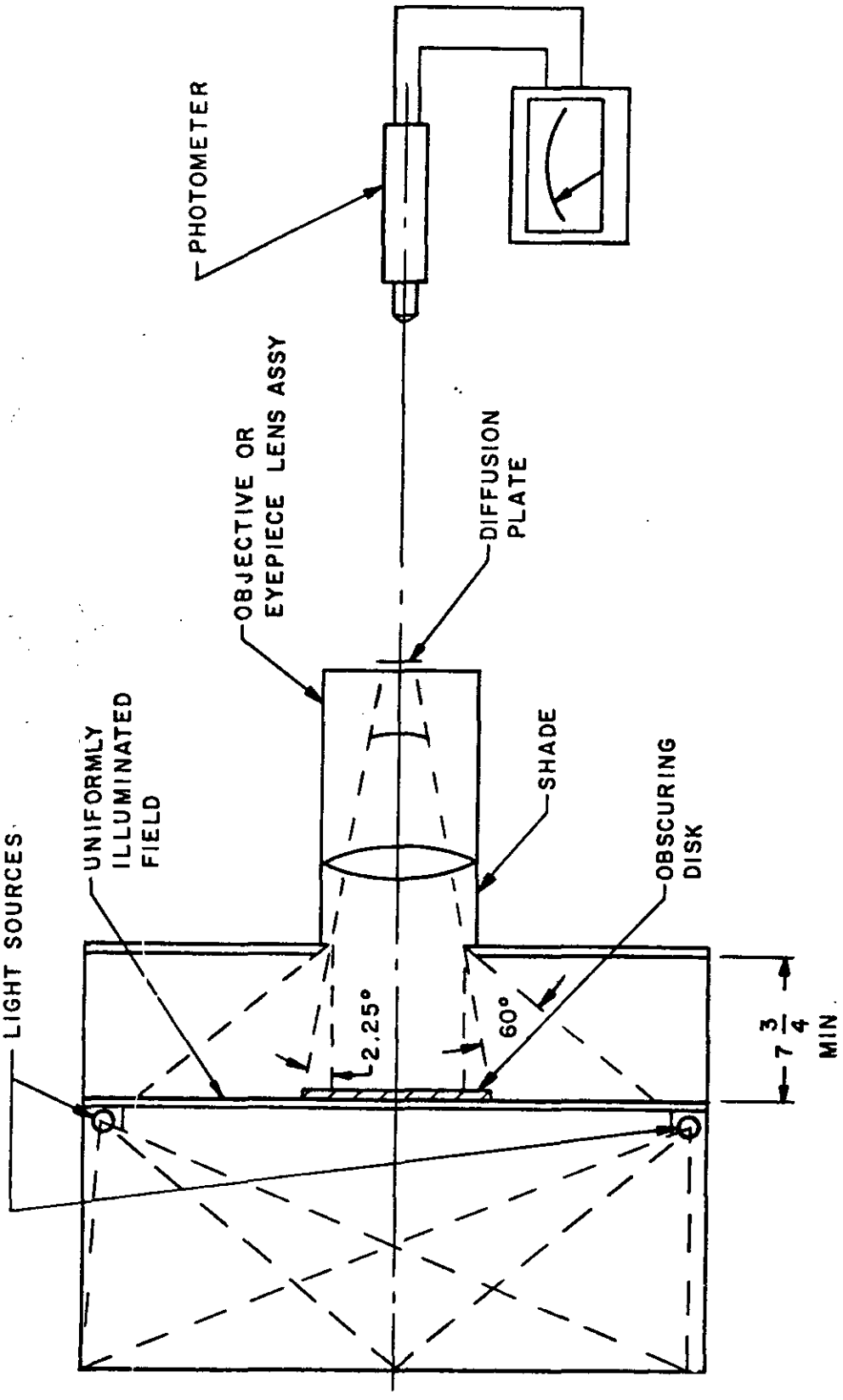


FIGURE 7

STRAY LIGHT AND TRANSMISSION TEST APPARATUS

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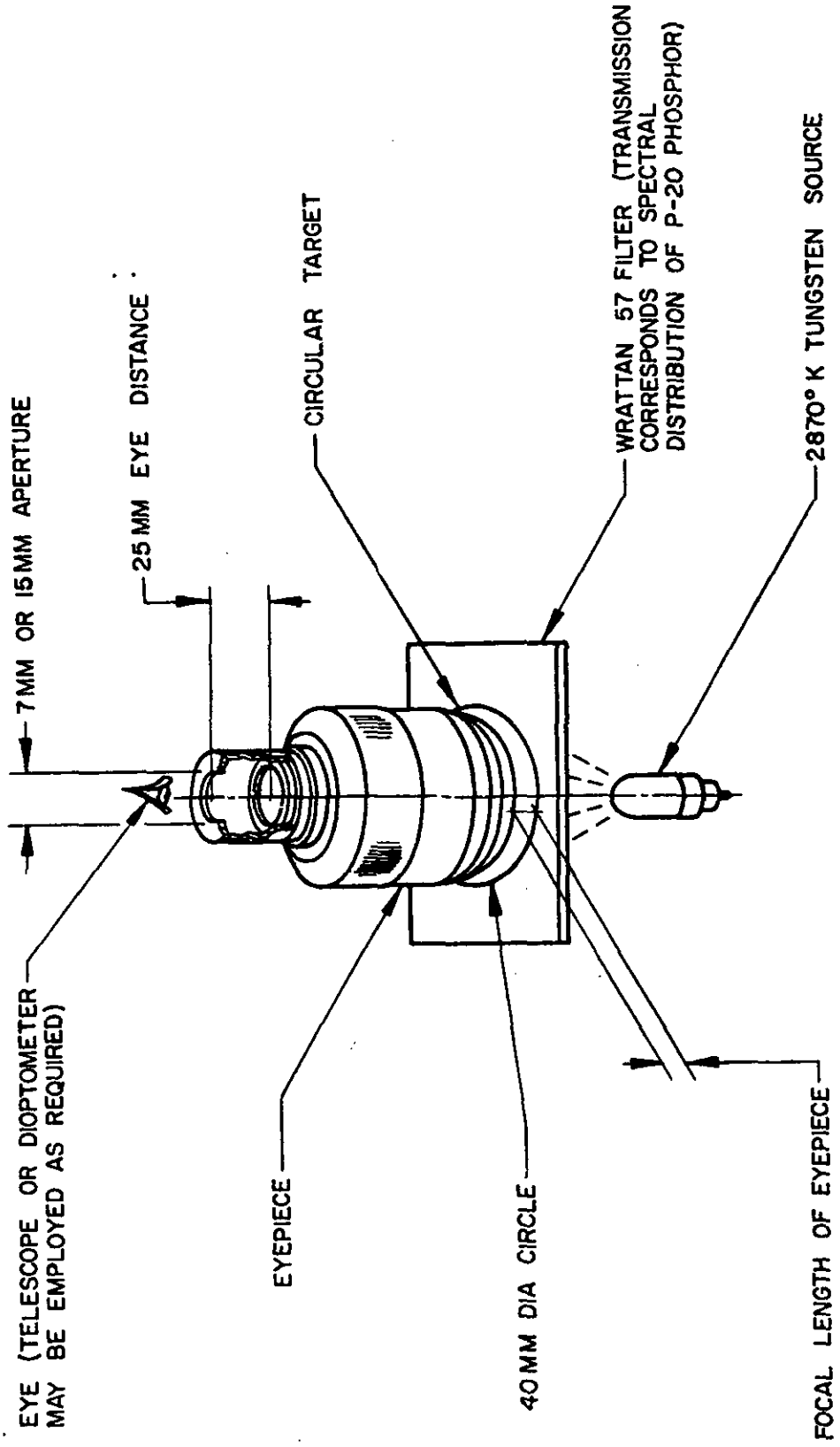


FIGURE 8

RESOLUTION, FIELD FLATNESS, EYE EXCURSION, FIELD COVERAGE AND EYE RELIEF DISTANCE TEST SCHEMATIC

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